

STOCK PRICE PREDICTION USING IBM SPSS MODELER





PROJECT OBJECTIVE

Goals of This Data Mining Project

- Analyze historical stock market data patterns
- Predict future stock prices with high accuracy
- Compare different prediction models to identify the best performer
- Leverage IBM SPSS Modeler as the primary data mining tool



DATASET ATTRIBUTES

Key Features of the Stock Market Dataset

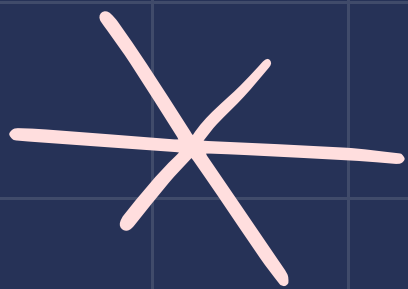
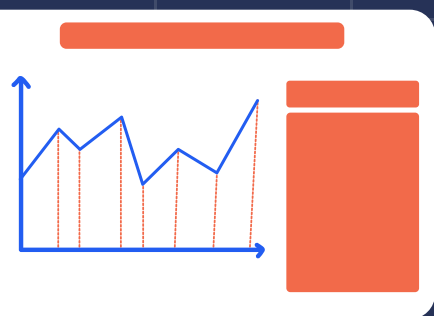
- Date: Trading day timestamp
- Open: Opening stock price of the day
- High: Highest price reached during the day
- Low: Lowest price reached during the day
- Close & Volume: Closing price and shares traded

WORKFLOW OVERVIEW

Key Steps in the Stock Price Prediction Process

- Import Dataset → Load stock market data into IBM SPSS Modeler
- Data Preprocessing → Handle missing values and normalize data
- Apply Auto Numeric Model → Automated modeling for price prediction
- Analyze Results → Evaluate model performance and accuracy
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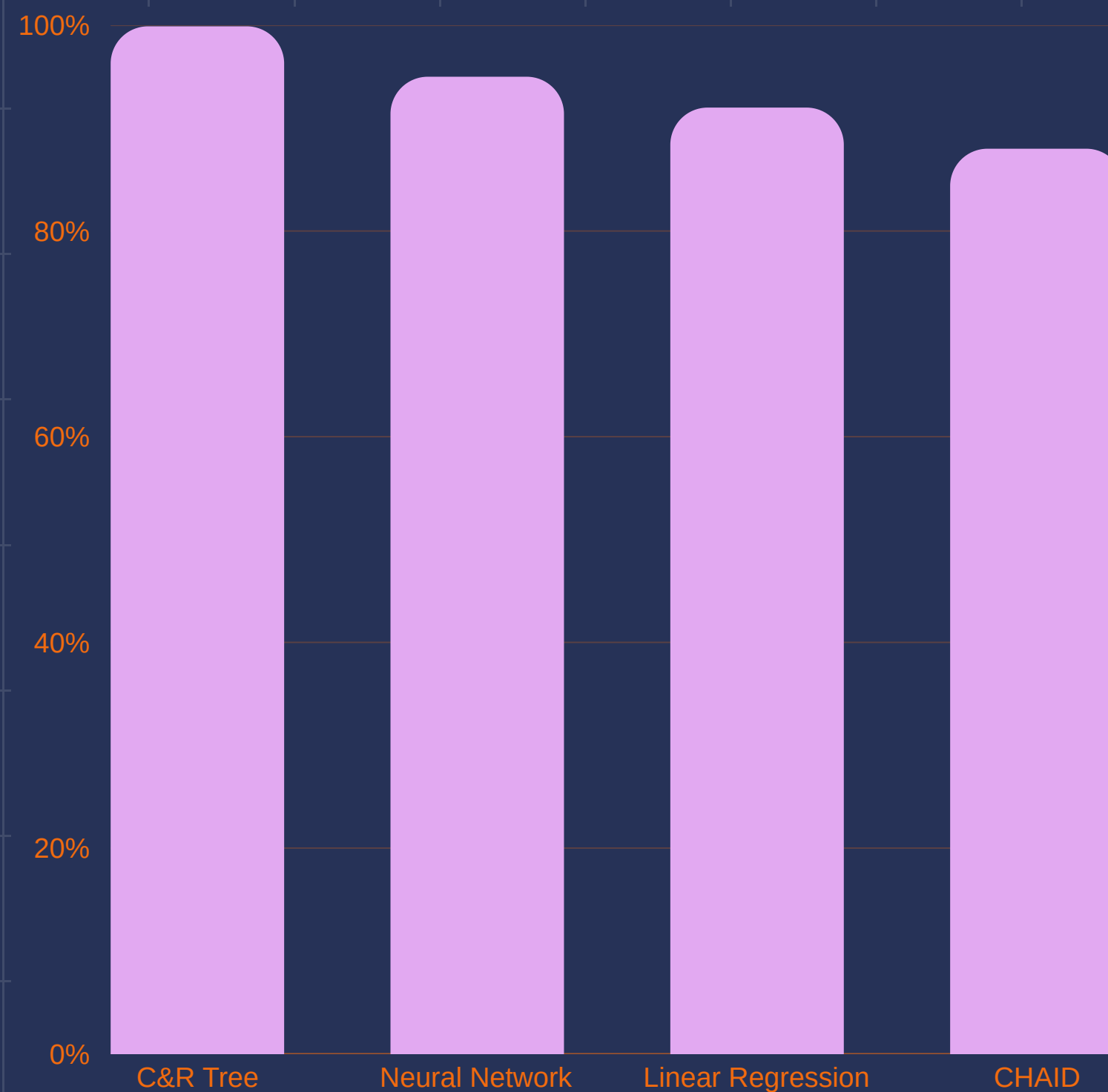




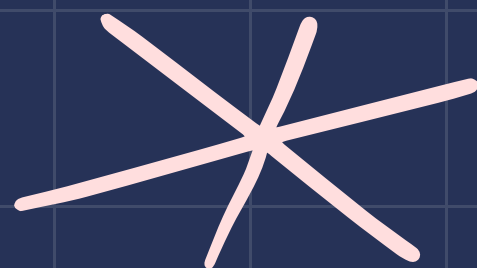
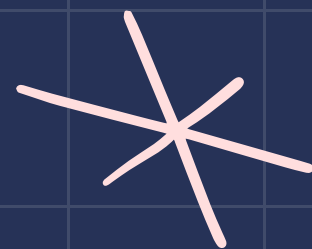
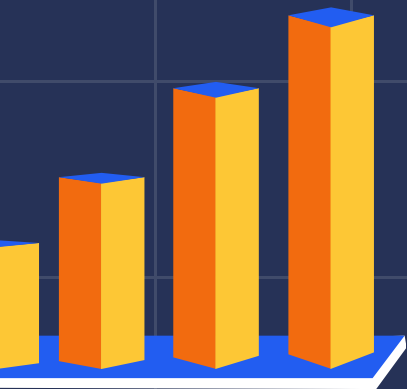
PREDICTION MODEL ACCURACY

Comparison of Prediction Model Accuracy

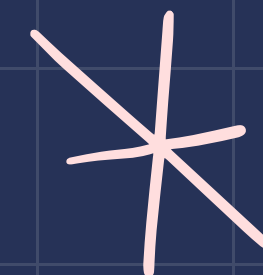
Our analysis evaluated multiple prediction models in IBM SPSS Modeler. The bar chart displays accuracy metrics across different algorithms, enabling clear comparison to identify the optimal model for stock price prediction.



Source: AI-generated data. Replace or verify before use.



BEST MODEL RESULTS



C&R Tree Model Top Performer

Correlation: 0.999 (near-perfect prediction).
Relative Error: 0.003 (minimal deviation).
C&R Tree outperforms other models due to its exceptional accuracy and reliability.



CONCLUSION

Key Takeaways from Our Analysis

- Successfully predicted stock prices using IBM SPSS Modeler
- C&R Tree model demonstrated outstanding accuracy (0.999 correlation)
- Approach applicable to other time series forecasting problems
- Data mining proves invaluable for financial market analysis
- Methodology can extend to other financial instruments

THANK YOU FOR YOUR ATTENTION

Questions & Discussion
Welcome

